

Nicholas Kern

NASA Hubble Fellow

CONTACT INFORMATION	University of Michigan Dept. of Physics Randall Laboratory 450 Church St. Ann Arbor, MI 48019	E-mail: nkern.astro@gmail.com Web: nkern.github.io ORCID: 0000-0002-8211-1892
EMPLOYMENT	NASA Hubble Fellow Department of Physics University of Michigan, Ann Arbor, MI, USA MIT Kavli Institute for Astrophysics and Space Research Massachusetts Institute of Technology, Cambridge, MA, USA Pappalardo Fellow MIT Kavli Institute for Astrophysics and Space Research Massachusetts Institute of Technology, Cambridge, MA, USA	Sept. 2024 – present Sept. 2023 – Sept. 2024 Sept. 2020 – Sept. 2023
EDUCATION	Ph.D., Astrophysics, University of California, Berkeley Advisor: Aaron R. Parsons M.A., Astrophysics, University of California, Berkeley B.S., Physics, Astrophysics, University of Michigan, Ann Arbor	August 2020 May 2017 May 2015
RESEARCH INTERESTS	cosmology, star and galaxy formation, ML/AI for science, data-intensive science, astrostatistics, radio astronomy techniques	
HONORS & AWARDS	Hubble Fellow, NASA Pappalardo Fellow, MIT, Department of Physics Mary Elizabeth Uhl Dissertation Prize, UC Berkeley, Department of Astronomy Teaching Effectiveness Award, UC Berkeley Outstanding Graduate Student Instructor Award, UC Berkeley Graduated with Highest Honors and Distinction, University of Michigan Excellence in Astrophysics Research Award, University of Michigan International Institute Fellow, University of Michigan Upper-Level Writing Prize in the Natural Sciences, University of Michigan	2023 2020 2020 2017 2017 2015 2015 2014 2014
STUDENTS ADVISED	- Jiaming Pan, U. Michigan PhD student <i>Exploring the generalizability of diffusion models for cosmology</i> - Prakhar Bansal, U. Michigan PhD student <i>Forecasting 21cm cosmology constraints on dark energy and fundamental physics</i> - Robert Pascua, McGill PhD Student <i>Quantifying signal reconstruction fidelity with radio telescope forward models</i> - Honggeun Kim, MIT PhD Student <i>Mitigating the impact of antenna feed deviations for 21 cm cosmology</i> - Ntsikelelo Charles, Rhodes U., South Africa, PhD student <i>Interferometric gain calibration in the presence of mutual coupling</i> <i>Mitigating diffuse foregrounds for interferometric calibration</i> - Cynthia De Los Santos, San Bernardino Community College <i>Spherical Harmonic Decomposition of HERA's Primary Beam</i>	Summer 2026 – present Summer 2025 – present Fall 2023 – Spring 2025 Fall 2021 – Fall 2024 Spring 2021 – Fall 2024 Summer 2022

	- Eleanor Rath, MIT PhD student Fall 2021 – Fall 2022 <i>A Bayesian framework for modeling antenna beam perturbations</i> - Duncan Rocha, Harvey Mudd undergrad (→ U. Chicago grad) Summer 2017 <i>Detectability of Alcock Paczynski effects for 21 cm intensity mapping</i> - Timothy Wilson, UCLA undergrad (→ UCLA grad) Summer 2016 <i>An MCMC sampler for semi-numerical Cosmic Dawn simulations</i>
TEACHING EXPERIENCE	- Session Instructor for <i>Interferometric Calibration and Imaging</i> 2018 – 2021 - Designed and taught a 3-hour lesson for the HERA summer undergraduate bootcamp - Head Instructor for <i>Python Programming in Astronomy</i> at UC Berkeley Summer 2017 - Developed a new 6-week intensive undergrad summer course for the Astronomy Dept. - Focus on scientific computing, computational physics, & data science - Lectured daily, held office hours, wrote and graded midterms, oversaw final projects - Graduate Instructor for <i>Introduction to Astrophysics</i> at UC Berkeley Fall 2016 - Led discussion section, developed interactive worksheets, graded homework & exams - Awarded department-wide “Outstanding Graduate Instructor” and university-wide “Teaching Effectiveness Award” - Graduate Instructor for <i>Stellar Structure & Evolution</i> at UC Berkeley Fall 2015 - Led discussion section, developed interactive worksheets, graded homework & exams - Undergraduate Instructor for <i>Introduction to Mechanics</i> at U. Michigan Spring 2015 - Taught undergraduates in breakout coding sessions, held office hours
SERVICE	To the Astrophysics Community: - Referee, Journal of Cosmology and Astroparticle Physics, 2025 – present - Referee, Radio Science 2020 – present - Referee, Monthly Notices of the Royal Astronomical Society 2019 – present - Referee, Astrophysical Journal 2018 – present At the University of Michigan - LOC for the 2025 Michigan Summer Cosmology School 2025 At the Massachusetts Institute of Technology - Primary Project Advisor, HERA Undergraduate Summer Research Bootcamp 2022 - Lead Coordinator, HERA Undergraduate Summer Research Bootcamp 2021 - Instructor & Mentor, HERA Undergraduate Summer Research Bootcamp 2020 – 2021 At the University of California, Berkeley - Graduate Representative, UC Berkeley Faculty Search Committee 2020 - Instructor & Mentor, HERA Undergraduate Summer Research Bootcamp 2017 – 2019 - Organizer, Astronomy Career Development Seminar 2016 – 2017 - Organizer, Graduate Student Colloquium Speaker Seminar 2015 – 2016
GRANTS AND COMPUTE ALLOCATIONS	Principal Investigator, Scaling-Up Differentiable Pipelines for Radio Surveys 2025 ACCESS, NCSA Delta Cluster Principal Investigator, Bayesian Frameworks for New 21 cm Telescopes 2022 XSEDE, PSC Bridges2 Cluster Principal Investigator, ML Tools for 21 cm Constraints on Fundamental Physics 2022 NERSC, Perlmutter Cluster
LEAD AUTHOR OR *STUDENT- ADVISED PUBLICATIONS	Lead author (9), *student-advised (5) 14. *Kim, H., ..., Kern, N., et al. 2025, <i>Exploring One-point Statistics in HERA Phase I Data: Effects</i>

of Foregrounds and Systematics on Measuring One-Point Statistics, [ApJ 993 189K](#)

13. **Kern, N.** 2025, *A Differentiable, End-to-End Forward Model for 21 cm Cosmology: Estimating the Foreground, Instrument, and Signal Joint Posterior*, [MNRAS 541 687K](#)
12. *Pascua, R., Martinot, Z., Liu, A., Aguirre, J., **Kern, N.**, et al. 2024, *A Generalized Method for Characterizing 21-cm Power Spectrum Signal Loss from Temporal Filtering of Drift-scanning Visibilities*, [ApJ 985 127P](#)
11. *Charles, N., **Kern, N.**, Pascua, R., et al. 2024, *Mitigating calibration errors from mutual coupling with time-domain filtering*, [MNRAS 534 3349C](#)
10. *Kim, H., **Kern, N.**, et al. 2023, *The Impact of Beam Variations on Power Spectrum Estimation for 21 cm Cosmology. II. Mitigation of Foreground Systematics for HERA*, [ApJ 953 136](#)
9. *Charles, N., **Kern, N.**, Bernardi, G. et al. 2023, *On the use of temporal filtering for mitigating galactic synchrotron calibration bias in 21 cm reionization observations*, [MNRAS 522 1009C](#)
8. Barry, N., Bernardi, G., Greig, B., **Kern, N.** (lead author) and Mertens, F. 2022, *SKA-Low Intensity Mapping Pathfinder Updates: Deeper 21 cm Power Spectrum Limits from Improved Analysis Frameworks*, [JATIS 8\(1\) 011007](#)
7. HERA Collaboration 2022, including **Kern, N.** (lead author), *First Results from HERA Phase I: Upper Limits on the Epoch of Reionization 21 cm Power Spectrum*, [ApJ 925 221A](#)
6. **Kern, N.** & Liu, A. 2021, *Gaussian Process Foreground Subtraction and Power Spectrum Estimation for 21 cm Cosmology*, [MNRAS 501 1463K](#)
5. **Kern, N.**, Dillon, J. S., Parsons, A. R., Carilli, C., Bernardi, G. et al. 2020, *Absolute Calibration Strategies for the Hydrogen Epoch of Reionization Array and Their Impact on the 21 cm Power Spectrum*, [ApJ 890 122](#)
4. **Kern, N.**, Parsons, A. R., Dillon, J. S., Lanman, A. E., et al. 2020, *Mitigating Internal Instrument Coupling for 21cm Cosmology. II. A Method Demonstration with the Hydrogen Epoch of Reionization Array*, [ApJ 888 70](#)
3. **Kern, N.**, Parsons, A. R., Dillon, J. S., Lanman, A. E., Fagnoni, N. and de Lera Acedo, E. 2019, *Mitigating Internal Instrument Coupling for 21cm Cosmology. I. Temporal and Spectral Modeling in Simulations*, [ApJ 884 105](#)
2. **Kern, N.**, Liu, A., Parsons, A. R., Mesinger, A., & Greig, B. 2017, *Emulating Simulations of Cosmic Dawn for 21 cm Power Spectrum Constraints on Cosmology, Reionization and X-ray Heating*, [ApJ 848 23](#)
1. **Kern, N. S.**, Keown, J. A., Tobin, J. J., Mead, A., & Gutermuth, R. 2016, *Radio Properties of Young Stellar Objects in the Serpens South Infrared Dark Cloud*, [AJ 151 42](#)

OTHER
PUBLICATIONS AS
CONTRIBUTING
AUTHOR

28. HERA Collaboration 2025 (including **Kern, N.**), *First Results from HERA Phase II*, submitted to [ApJ](#)
27. Chen, K., Wilensky, M., Liu, A., ..., **Kern, N.** et al. 2025, *Impacts and Statistical Mitigation of Missing Data on the 21cm Power Spectrum: A Case Study with the Hydrogen Epoch of Reionization Array*, [ApJ 979 191C](#)
26. Rath, E., Pascua, R., Josaitis, A., ..., **Kern, N.** et al. 2025, *Investigating Mutual Coupling in the Hydrogen Epoch of Reionization Array and Mitigating its Effects on the 21-cm Power Spectrum*, [Submitted to MNRAS](#)
25. Berkhout, L., Jacobs, D., ..., **Kern, N.**, et al. 2024, *Hydrogen Epoch of Reionization Array (HERA) Phase II Deployment and Commissioning*, [PASP 136d 5002B](#)

24. Murphy, G., Bull, P., Santos, M., ..., **Kern, N.** et al. 2024, *Bayesian estimation of cross-coupling and reflection systematics in 21cm array visibility data*, [MNRAS 534 2653M](#)
23. Keller, P., Nikolic, B., Thyagarajan, N., ..., **Kern, N.**, et al. 2023, *Search for the Epoch of Reionisation with HERA: Upper Limits on the Closure Phase Delay Power Spectrum*, [MNRAS 524 583K](#)
22. HERA Collaboration 2023, including **Kern, N.**, *Improved Constraints on the 21 cm EoR Power Spectrum and the X-Ray Heating of the IGM with HERA Phase I Observations*, [ApJ 945 124](#)
21. Pagano, M., Liu, J., Liu, A., **Kern, N.** et al. 2022, *Characterization Of Inpaint Residuals In Interferometric Measurements of the Epoch Of Reionization*, [MNRAS 520 5552P](#)
20. Kim, H., Nhan, B., Hewitt, J., **Kern, N.** et al. 2022, *The Impact of Beam Variations on Power Spectrum Estimation for 21-cm Cosmology I: Simulations of Foreground Contamination for HERA*, [ApJ 941 207](#)
19. Xu, Z., Hewitt, J., ..., **Kern, N.** et al. 2022, *Direct Optimal Mapping for 21cm Cosmology: A Demonstration with the Hydrogen Epoch of Reionization Array*, [ApJ 938 128](#)
18. HERA Collaboration 2022, including **Kern, N.**, *HERA Phase I Limits on the Cosmic 21-cm Signal: Constraints on Astrophysics and Cosmology During the Epoch of Reionization*, [ApJ 924 51A](#)
17. Aguirre, J., Murray, S., ..., **Kern, N.**, et al. 2021, *Validation of the HERA Phase I Epoch of Reionization 21 cm Power Spectrum Software Pipeline*, [ApJ 924 85A](#)
16. LaPlante, P., Williams, P. K. G., ..., **Kern, N.**, et al. 2021, *A Real Time Processing System for Big Data in Astronomy: Applications to HERA*, [A&C 3600489L](#)
15. Tan, J., Liu, A., **Kern, N.**, et al. 2021, *Methods of Error Estimation for Delay Power Spectra in 21cm Cosmology*, [ApJS 255 26T](#)
14. Ewall-Wice, A., **Kern, N.**, Dillon, J. S., et al. 2021, *DAYENU: A Simple Filter of Smooth Foregrounds for Intensity Mapping Power Spectra*, [MNRAS 500 5195E](#)
13. Nunhokee, C. D., Parsons, A. R., **Kern, N.**, et al. 2020, *Measuring HERA's primary beam in-situ: methodology and first results*, [ApJ 897 5N](#)
12. Thyagarajan, N., Carilli, C., Nikolic, B., ..., **Kern, N.**, et al. 2020, *Detection of Cosmic Structures using the Bispectrum Phase. II. First Results from Application to Cosmic Reionization Using the Hydrogen Epoch of Reionization Array*, [Phys. Rev. D 102, 022002](#)
11. Dillon, J. S., Lee, M., Ali, Z. S., ..., **Kern, N.**, et al. 2020, *Redundant-Baseline Calibration of the Hydrogen Epoch of Reionization Array*, [MNRAS 499 5840D](#)
10. Ghosh, A., Mertens, F., Bernardi, G., ..., **Kern, N.**, et al. 2020, *Foreground modelling via Gaussian process regression: an application to HERA data*, [MNRAS 495 2813G](#)
9. Carilli, C., Thyagarajan, N., Kent, J., ..., **Kern, N.**, et al. 2020, *Imaging and Modeling Data from the Hydrogen Epoch of Reionization Array*, [ApJS 247 67](#)
8. Lanman, A. E., Poher, J. C., **Kern, N.**, et al. 2020, *Quantifying EoR delay spectrum contamination from diffuse radio emission*, [MNRAS 494 3712L](#)
7. Monsalve, R. A., Greig, B., Bowman, J. D., ..., **Kern, N.**, et al. 2018, *Results from EDGES High-Band: II. Constraints on Parameters of Early Galaxies*, [ApJ 863 11](#)
6. Kohn, S. A., Aguirre, J. E., La Plante, P., ..., **Kern, N.**, et al. 2018, *The HERA-19 Commissioning Array: Direction Dependent Effects*, [ApJ 882 58K](#)
5. Dillon, J. S., Kohn, S. A., Parsons, A. R., ..., **Kern, N.**, et al. 2017, *Polarized redundant-baseline calibration for 21 cm cosmology without adding spectral structure*, [MNRAS 477 5670](#)
4. Miller, C. J., Stark, A., Gifford D., & **Kern, N.** 2016, *Inferring Gravitational Potentials from Mass Densities in Cluster-Sized Halos*, [ApJ 822 41](#)

3. Stark, A., Miller, C. J., **Kern, N.**, Gifford, D., et al. 2016, *Probing Theories of Gravity with Phase Space-Inferred Potentials of Galaxy Clusters*, [Phys. Rev. D 93, 084036](#)
2. Gifford, D., **Kern, N.**, & Miller, C. 2016, *Stacking Caustic Masses from Galaxy Clusters*, [ApJ 834 204](#)
1. Gifford, D., Miller, C. J., & **Kern, N.** 2013, *A Systematic Analysis of Caustic Methods for Galaxy Cluster Masses*, [ApJ 773 116](#)

SELECTED TALKS	Minnesota Institute for Astronomy (MIfA) Colloquium Minneapolis, MN	February 2026
	U. Toledo Special Colloquium Toledo, OH	February 2026
	Kavli Institute for Theoretical Physics Program on Multi-Probe Cosmology Santa Barbara, CA	February 2026
	Michigan AI Summit Research Vision Talk Ann Arbor, MI	November 2025
	NASA Hubble Symposium Baltimore, MD	October 2025
	U. Toledo Astronomy Colloquium Toledo, OH	September 2025
	Michigan HEP-Astro Seminar Ann Arbor, MI	September 2025
	UIUC Astrophysics & Cosmology Seminar Urbana-Champaign, IL	September 2025
	UIUC Astronomy Colloquium Urbana-Champaign, IL	September 2025
	Wisconsin Cosmology Seminar Madison, WI	May 2025
	Line Intensity Mapping Conference Heidelberg, Germany	December 2024
	NASA Hubble Symposium Los Angeles, CA	September 2024
	Johns Hopkins Theoretical Cosmology Seminar Baltimore, MD	December, 2023
	NASA Hubble Symposium Cambridge, MA	September, 2023
	Wisconsin 21 cm Workshop Madison, Wisconsin	August, 2022
	3rd URSI Atlantic Radio Science Meeting Gran Canaria, Spain	June 2022
	ASTRON/JIVE Colloquium ASTRON, Dwingeloo, Netherlands	March 2022
	Royal Astronomical Society Specialist Discussion Virtual	December 2021
	Science at Low Frequencies VIII Virtual	December 2021

Astrophysics Seminar Johns Hopkins University, Baltimore, MA	November 2021
INAF Joint Astrophysical Colloquium INAF, Bologna, Italy	November 2021
Pappalardo Research Symposium MIT, Cambridge, MA, USA	May 2021
A Precursor View of the SKA Sky Virtual	March 2021
Observing the First Billion Years IIT Indore, Indore, India	January 2020
Observational Cosmology Seminar California Institute of Technology, Pasadena, CA	December 2019
Center for Astrophysics SMA Seminar Center for Astrophysics, Cambridge, MA, USA	November 2019
MIT Kavli Institute Brown Bag Lunch Talks MIT, Cambridge, MA, USA	November 2019
BCCP Cosmology Workshop University of California, Berkeley, CA, USA	January 2018
JILA Astrophysics Seminar University of Colorado, Boulder, CO, USA	October 2017
NASA Machine Learning Workshop Mountain View, CA, USA	August 2017